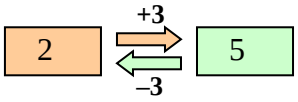
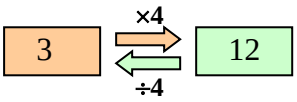


Determinant and Inverse Matrices
(行列式與逆矩陣)

There are a number of inverse operations (逆運算) in mathematics, such as



Inverse of addition is subtraction



Inverse of multiplication is division

Inverse of addition (i.e. subtraction) is well defined in matrices:

$$\begin{pmatrix}3&2&5\\4&3&2\end{pmatrix} \xrightarrow{+\begin{pmatrix}1&1&1\\2&2&2\end{pmatrix}} \begin{pmatrix}4&3&6\\6&5&4\end{pmatrix} \xrightarrow{-\begin{pmatrix}1&1&1\\2&2&2\end{pmatrix}} \begin{pmatrix}3&2&5\\4&3&2\end{pmatrix}$$

However, inverse of multiplication (i.e. division) is not defined in matrices!

$$\begin{pmatrix}1&2\\3&4\end{pmatrix} \xrightarrow{\times\begin{pmatrix}5&7\\6&8\end{pmatrix}} \begin{pmatrix}17&23\\39&53\end{pmatrix} \xrightarrow{\div\begin{pmatrix}5&7\\6&8\end{pmatrix}} \begin{pmatrix}1&2\\3&4\end{pmatrix}$$



To perform inverse of multiplication in matrices, we need to make use of **Inverse Matrices (逆矩陣).**

For example, consider $A=\begin{pmatrix}1&3\\2&5\end{pmatrix}$, then its inverse (denoted as A^{-1}), if exists,

can be expressed as $A^{-1}=\begin{pmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{pmatrix}$, where $AA^{-1}=A^{-1}A=I$

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Concepts of Inverse Matrices (逆矩陣之概念)

Definition: Inverse of Matrix (逆矩陣之定義)

Let A be a square matrix. If B is a square matrix such that

$AB = BA = I$ where I is the identity matrix

then B is callled the ***inverse*** of A and we write $B = A^{-1}$. Also, $A = B^{-1}$.

If a matrix has an inverse, then it is ***non-singular*** **(非奇異方陣)**

Otherwise, it is ***singular*** **(奇異方陣).**

Examples:

- $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$, $B = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}$ are inverse matrix of each other, It can see that

$$AB = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}\begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad BA = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

i.e. $B = A^{-1}$ also $A = B^{-1}$.

Therefore A and B are both **non-singular.**

- $C = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$ does not have an inverse, because if we let $X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}$,

$$CX = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}\begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} = \begin{pmatrix} x_{11} + 2x_{21} & x_{12} + 2x_{22} \\ 3x_{11} + 6x_{21} & 3x_{12} + 6x_{22} \end{pmatrix}$$

We can see that $CX = \begin{pmatrix} x_{11} + 2x_{21} & x_{12} + 2x_{22} \\ 3(x_{11} + 2x_{21}) & 3(x_{12} + 2x_{22}) \end{pmatrix}$ which is impossible

to be equal to identity matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ for any values of x_{mn}.

Therefore C is **singular**.



Concepts of Inverse Matrices

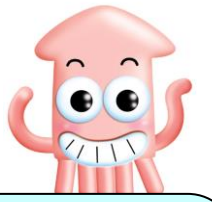
Remarks:

1. 'A is non-singular (非奇異方陣)' can be written as 'A is ***invertible* (可逆方陣)'.**

2. **Division is undefined** for matrix, and hence, we should **NOT** use the symbol "$\frac{1}{A}$" to represent A^{-1}.

3. $I^{-1} = I$ i.e. the inverse of I is I itself.

4. The inverse of **O** = $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ does not exist, i.e 0 is **singular**.



Theorem: Properties of Multiplicative Inverse (逆矩陣之乘法特性)

Let A and B be two **non-singular** matrices and α be a scalar, n be a positive integer. Then

(1) The inverse of A is unique (唯一) and denote as A^{-1}.

(2) $AA^{-1} = A^{-1}A = I$

(3) $(A^{-1})^{-1} = A$

(4) $(\alpha A)^{-1} = \frac{1}{\alpha} A^{-1}$

(5) $(AB)^{-1} = B^{-1}A^{-1}$

(6) $(A^n)^{-1} = (A^{-1})^n$

(7) $(A^T)^{-1} = (A^{-1})^T$

Concepts of Inverse Matrices

[M2 2018 Q.7]

Define $M = \begin{pmatrix} 7 & 3 \\ -1 & 5 \end{pmatrix}$. Let $X = \begin{pmatrix} a & 6a \\ b & c \end{pmatrix}$ be a matrix such that $MX = XM$.

(a) Express b and c in terms of a.

(b) Prove that X is a non-singular matrix.

(c) Express $(X^T)^{-1}$ in terms of a. ***(8 marks)***

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Concepts of Inverse Matrices

[P.Math 1996 Paper I Q.1]

Let $A=\begin{pmatrix}0 & 0 & -2\\ 1 & 2 & 1\\ 1 & 0 & 3\end{pmatrix}$.

(a) Evaluate $A^3-5A^2+8A-4I$.

(b) Hence, or otherwise, find A^{-1}. ***(6 marks)***

註： 上題為 Cayley-Hamilton Theorem 的其中一個應用，惟這不會在單元二課程中探討。



Determinant of Matrices (矩陣之行列式)

To find the inverse of a matrix A (i.e. A^{-1}), we need to ensure that it does exist. In other words, we need to show that A is **non-singular** first. It can be done by calculating the **Determinant (行列式)** of the matrix A.

Definition: Determinant of square matrix with order 2 (2 階方陣之行列式)

For any 2×2 square matrix $A=\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, the ***determinant*** of A, denoted by

$\det A$ or $|A|$ can be found by

$$|A| = \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Example: If $A = \begin{pmatrix} 3 & -1 \\ 5 & 2 \end{pmatrix}$, then $\det A = \begin{vmatrix} 3 & -1 \\ 5 & 2 \end{vmatrix} = (3)(2) - (-1)(5) = 11$

More Example:

Let $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \mathbf{R}$, $a \neq 0$ and $\det A = 0$.

Show that $A=\begin{pmatrix} a & b \\ ka & kb \end{pmatrix}$ for some $k \in \mathbf{R}$.

Solution

$\because \quad \det A = 0$

$\therefore \quad ad - bc = 0$

$ad = bc$

Hence, by putting $k = \dfrac{c}{a}$, we have

$c = ka$ and $d = \dfrac{bc}{a} = kb$ i.e. $A = \begin{pmatrix} a & b \\ ka & kb \end{pmatrix}$.

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Determinant of Matrices



[P.Math 1996 Paper I Q.8(a)]

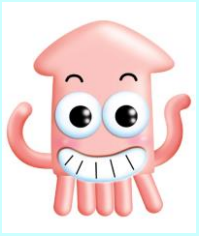
Solve the equation $\det\left[\begin{pmatrix}1 & 3\\ 2 & 2\end{pmatrix}-\lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\right]=0$. ***(3 marks)***

Determinant of Matrices

Theorem:

Let A be a square matrix of any order.

- A is non-singular if and only if $\det A \neq 0$.
- A is singular if and only if $\det A = 0$.



Examples:

- Let $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$, then $\det A = \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = (1)(5) - (3)(2) = -1$.

  $\therefore$ A is non-singular and A^{-1} exist.

- Let $B = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}$, then $\det B = \begin{vmatrix} -5 & 3 \\ 2 & -1 \end{vmatrix} = (-5)(-1) - (3)(2) = -1$.

  $\therefore$ B is non-singular and B^{-1} exist.

- Let $C = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$, then $\det C = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = (1)(6) - (2)(3) = 0$.

  $\therefore$ C is singular and C^{-1} does not exist.

Remarks:

In general, if $M = \begin{pmatrix} a & b \\ ka & kb \end{pmatrix}$, then $\det M = 0$ and M^{-1} does not exist.

Proof:

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Determinant of Matrices (order 3)

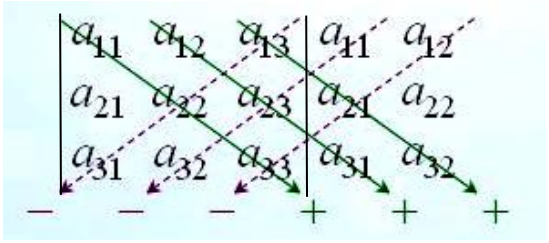
Definition: Determinant of square matrix with order 3 (3 階方陣之行列式)

For any 3×3 square matrix $A=\left[a_{ij}\right]_{3\times3}$

$$|A|=\det A=\begin{vmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{vmatrix}$$

$$=a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32}-a_{11}a_{23}a_{32}-a_{12}a_{21}a_{33}-a_{13}a_{22}a_{31}$$

Note: The determinant can be found by "Rule of Sarrus" as follows:



由左上向右下相乘 ➔ 加
由右上向左下相乘 ➔ 減

Example:

$$\begin{vmatrix}1&-2&3\\5&6&8\\-4&9&7\end{vmatrix}=(1)(6)(7)+(-2)(8)(-4)+(3)(5)(9)-(1)(8)(9)-(-2)(5)(7)-(3)(6)(-4)$$

$$=42+64+135-72+70+72$$

$$=311$$

$$\begin{vmatrix}1&2&3\\3&6&9\\-4&8&-7\end{vmatrix}=(1)(6)(-7)+(2)(9)(-4)+(3)(3)(8)-(1)(9)(8)-(2)(3)(-7)-(3)(6)(-4)$$

$$=-42-72+72-72+42+72$$

$$=0$$

Remarks:

In general, if $M=\begin{pmatrix}a&b&c\\ka&kb&kc\\?&?&?\end{pmatrix}$, then $\det M = 0$ and M^{-1} does not exist.

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Determinant of Matrices (order n)

For square matrices with order higher than 3, the Rule of Sarrus no longer applies. So some mathematicians have found an alternative way to calculate the determinant:

Definition: Determinant of square matrix with order *n* (*n* 階方陣之行列式)

For any $n \times n$ square matrix $M = \left[a_{ij} \right]_{n \times n}$,

$$|M| = \det M = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$= a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + ... + a_{1n}A_{1n}$ (expand along 1st row)

OR $= a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} + ... + a_{2n}A_{2n}$ (expand along 2nd row)

OR $= a_{n1}A_{n1} + a_{n2}A_{n2} + a_{n3}A_{n3} + ... + a_{nn}A_{nn}$ (expand along nth row)

OR $= a_{1n}A_{1n} + a_{2n}A_{2n} + a_{3n}A_{3n} + ... + a_{nn}A_{nn}$ (expand along nth row column)

where A_{ij} is the **cofactor (餘因式)** of the element a_{ij} in M and

$$A_{ij} = (-1)^{i+j} \cdot \det \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2j} & \cdots & a_{2n} \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdots & \cdot \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{ij} & \cdots & a_{in} \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdots & \cdot \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nj} & \cdots & a_{nn} \end{pmatrix}$$

註：餘因式 (cofactor) 的概念十分重要，請同學務必緊記！

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Determinant of Matrices (order n)

Example: Find $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$.

Solution

Expand along the 1st row:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = (1)(-1)^{1+1}\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} + (2)(-1)^{1+2}\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + (3)(-1)^{1+3}\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= [(5)(9) - (6)(8)] - 2[(4)(9) - (6)(7)] + 3[(4)(8) - (5)(7)]$$

$$= 0$$

OR Expand along the 2nd row:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = (4)(-1)^{2+1}\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} + (5)(-1)^{2+2}\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} + (6)(-1)^{2+3}\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$$

$$= -4[(2)(9) - (3)(8)] + 5[(1)(9) - (3)(7)] - 6[(1)(8) - (2)(7)]$$

$$= 0$$

OR Expand along the 3rd column:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = (3)(-1)^{1+3}\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} + (6)(-1)^{2+3}\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} + (9)(-1)^{3+3}\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}$$

$$= 3[(4)(8) - (5)(7)] - 6[(1)(8) - (2)(7)] + 9[(1)(5) - (2)(4)]$$

$$= 0$$

由於課程只包括 2 階及 3 階方陣的行列式運算，所以在此不示範 4 階或以上方陣的行列式運算，同學有空時可自行嘗試運算並多加探究。

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Properties of Determinant (行列式之特性)

Theorem: Basic Properties of Determinants

Let $A=\left[a_{ij}\right]_{n\times n}$, $B=\left[b_{ij}\right]_{n\times n}$ and k be a scalar. Then

(1) $\det A^T = \det A$.

(2) $\det(A^n) = (\det A)^n$

(3) $\det(AB) = (\det A)(\det B)$

(4) $\det(kA) = k^n \det A$

(5) $\det I = 1$

(6) If A is a non-singular, then $\det A \neq 0$ and $\det A^{-1} = \dfrac{1}{\det A}$.

More Example:

Let A and C be two square matrix of the same order. Show that $\det(ACA^{-1}) = \det C$.

Solution

$$\det(ACA^{-1})$$

$$= (\det A)(\det C)(\det A^{-1})$$

$$= (\det C)\left[(\det A)(\det A^{-1})\right]$$

$$= (\det C)(\det AA^{-1})$$

$$= (\det C)(\det I)$$

$$= \det C$$

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Properties of Determinant

[M2 2015 Q.6]

(a) Let M be a 3×3 real matrix such that $M^T = -M$, where M^T is the transpose of M. Prove that $|M| = 0$.

(b) Let $A = \begin{pmatrix} -1 & a & b \\ -a & -1 & -8 \\ -b & 8 & -1 \end{pmatrix}$, where a and b are real numbers. Denote the 3×3 identity matrix by I.

(i) Using (a), or otherwise, prove that $|A + I| = 0$.

(ii) Someone claims that $A^3 + I$ is a singular matrix. Do you agree? Explain your answer.

(6 marks)

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Properties of Determinant

[P.Math 1993 Paper I Q.6]

(a) Show that if A is a 3×3 matrix such that $A^t = -A$, then $\det A = 0$.

(b) Given that $B = \begin{pmatrix} 1 & -2 & 74 \\ 2 & 1 & -67 \\ -74 & 67 & 1 \end{pmatrix}$

Use (a), or otherwise,to show $\det(I - B) = 0$.

Hence deduce that $\det(I - B^4) = 0$. ***(7 marks)***

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Computation of Inverse Matrix (計算逆矩陣)

Theorem: Inverse of a Matrix by Determinants (用行列式計算逆矩陣)

Let $M = \left[a_{ij} \right]_{n \times n}$ be a square matrix. If $\det M \neq 0$, then

$$M^{-1} = \frac{1}{\det M} \operatorname{adj} M$$

where adj M is the **adjoint matrix (伴隨矩陣)** of M and

$$\operatorname{adj} M = \left[A_{ij} \right]_{n \times n}^{T} = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{pmatrix}^{T}$$

and A_{ij} is the **cofactor (餘因式)** of the element a_{ij} in M and

$$A_{ij} = (-1)^{i+j} \cdot \det \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2j} & \cdots & a_{2n} \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdots & \cdot \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{ij} & \cdots & a_{in} \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdots & \cdot \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nj} & \cdots & a_{nn} \end{pmatrix}$$

註： 計算逆矩陣的方法有很多種。除了用行列式去計算外，於下一章我們亦會探討如何用高斯消去法 (Gaussian elimination) 去計算逆矩陣。

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Computation of Inverse Matrix (計算逆矩陣)

Example: Let $A=\begin{pmatrix}3&2&6\\1&1&2\\2&2&5\end{pmatrix}$. Find A^{-1}.

Solution

By expanding along the first row, we have

$$\det A=(3)(-1)^{1+1}\begin{vmatrix}1&2\\2&5\end{vmatrix}+(2)(-1)^{1+2}\begin{vmatrix}1&2\\2&5\end{vmatrix}+(6)(-1)^{1+3}\begin{vmatrix}1&1\\2&2\end{vmatrix}$$

$$=3(5-4)-2(5-4)+6(2-2)$$

$$=1$$

Hence A is non-singular.

Step 1: 計算 det A 並證明 A 是非奇異矩陣 **non-singular** (det $A\neq 0$)

$$\text{adj}\,A=\begin{pmatrix}\begin{vmatrix}1&2\\2&5\end{vmatrix}&-\begin{vmatrix}1&2\\2&5\end{vmatrix}&\begin{vmatrix}1&1\\2&2\end{vmatrix}\\-\begin{vmatrix}2&6\\2&5\end{vmatrix}&\begin{vmatrix}3&6\\2&5\end{vmatrix}&-\begin{vmatrix}3&2\\2&2\end{vmatrix}\\\begin{vmatrix}2&6\\1&2\end{vmatrix}&-\begin{vmatrix}3&6\\1&2\end{vmatrix}&\begin{vmatrix}3&2\\1&1\end{vmatrix}\end{pmatrix}^T$$

$$=\begin{pmatrix}1&-1&0\\2&3&-2\\-2&0&1\end{pmatrix}^T$$

$$=\begin{pmatrix}1&2&-2\\-1&3&0\\0&-2&1\end{pmatrix}$$

Step 2: 計算 adj A : 先計算 cofactor，然後再轉置 (transpose)

$$\therefore\quad A^{-1}=\frac{1}{\det A}\text{adj}\,A=\frac{1}{1}\begin{pmatrix}1&2&-2\\-1&3&0\\0&-2&1\end{pmatrix}$$

$$=\begin{pmatrix}1&2&-2\\-1&3&0\\0&-2&1\end{pmatrix}$$

Step 3: 最後計算 A^{-1} :

$$A^{-1}=\frac{1}{\det A}\text{adj}\,A$$

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Computation of Inverse Matrix

[M2 2013 Paper I Q.8]

Let M be the matrix $\begin{pmatrix} 1 & k & 0 \\ 0 & 1 & 1 \\ k & 0 & 0 \end{pmatrix}$, where $k \neq 0$.

(a) Find M^{-1}.

(b) If $M \begin{pmatrix} x \\ 1 \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$, find the value of k. ***(5 marks)***

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n^{th} Power of Square Matrix (by diagonization)



還記得早前介紹計算矩陣 A 之冪 (次方) 的方法嗎？

Method 1: 計算 $A^2, A^3, \ldots$ 並尋中箇中規律，從而推測 A^n

Method 2: 運用數學歸納法 (Mathematical Induction)

Method 3: 矩陣對角化 (Diagonalization)

若果任何方陣 A 及 B，$B^{-1}AB$ 是對角矩陣的話，我們就可以用矩陣對角法 (Diagonalization) 去計算 A^n：

Example

Let $A=\begin{pmatrix}1 & 0\\ 1 & 3\end{pmatrix}$, $B=\begin{pmatrix}-2 & 0\\ 1 & \lambda\end{pmatrix}$.

(a) If B^{-1} exists and $B^{-1}AB=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$, find λ, a and b.

(b) Hence find A^{100}.

Solution (a) $\because \quad \det B=-2\lambda$

$$\therefore \quad B^{-1}=\frac{1}{-2\lambda}\begin{pmatrix}\lambda & 0\\ -1 & -2\end{pmatrix}$$

Hence, $B^{-1}AB=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$

$$\frac{1}{-2\lambda}\begin{pmatrix}\lambda & 0\\ -1 & -2\end{pmatrix}\begin{pmatrix}1 & 0\\ 1 & 3\end{pmatrix}\begin{pmatrix}-2 & 0\\ 1 & \lambda\end{pmatrix}=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$$

$$\frac{1}{-2\lambda}\begin{pmatrix}\lambda & 0\\ -3 & -6\end{pmatrix}\begin{pmatrix}-2 & 0\\ 1 & \lambda\end{pmatrix}=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$$

$$\frac{1}{-2\lambda}\begin{pmatrix}-2\lambda & 0\\ 0 & -6\lambda\end{pmatrix}=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$$

$$\begin{pmatrix}1 & 0\\ 0 & 3\end{pmatrix}=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$$

$$\therefore \quad a=1, b=3.$$

As B^{-1} exist, $\det B\neq 0$, λ can be any real number except 0.

Not for sale

n^{th} Power of Square Matrix (by diagonization)

(b) From (a), $B^{-1}AB = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ and so,

$$(B^{-1}AB)^{100} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}^{100}$$

$$B^{-1}A^{100}B = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}^{100}$$

$$B(B^{-1}A^{100}B)B^{-1} = B\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}^{100} B^{-1}$$

$$A^{100} = B\begin{pmatrix} 1^{100} & 0 \\ 0 & 3^{100} \end{pmatrix} B^{-1}$$

Then, by putting $\lambda = 1$, we have

$$B = \begin{pmatrix} -2 & 0 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad B^{-1} = \frac{1}{-2}\begin{pmatrix} 1 & 0 \\ -1 & -2 \end{pmatrix}$$

and so,

$$A^{100} = -\frac{1}{2}\begin{pmatrix} -2 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 3^{100} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -1 & -2 \end{pmatrix}$$

$$= -\frac{1}{2}\begin{pmatrix} -2 & 0 \\ 1 & 3^{100} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -1 & -2 \end{pmatrix}$$

$$= -\frac{1}{2}\begin{pmatrix} -2 & 0 \\ 1-3^{100} & -2\times 3^{100} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ \dfrac{3^{100}-1}{2} & 3^{100} \end{pmatrix}$$

Technique !!!

若果方陣 A 及 B 可寫成 $B^{-1}AB = D$，其中 D 是對角方陣

則

$(B^{-1}DB)^n$

$= (B^{-1}DB)\,(B^{-1}DB)\ldots\,(B^{-1}DB)$

$= B^{-1}DBB^{-1}DB \ldots B^{-1}DB$

$= B^{-1}DDD \ldots DB$

$= B^{-1}D^nB$

而 D^n 是十分容易計算的。

註： 要知道某個方陣 A 是否能用矩陣對角化去計算 A^n，我們可以從 A 的特徵值 (eigenvalue) 去判斷，惟這不會在單元二課程中探討。

n^{th} Power of Square Matrix (by diagonization)

[M2 2012 Paper I Q.11]

(a) Solve the equation

$$\begin{vmatrix} 1-x & 4 \\ 2 & 3-x \end{vmatrix} = 0 \text{ ----- (*).}$$ ***(2 marks)***

(b) Let x_1, x_2 $(x_1 < x_2)$ be the roots of (*). Let $P = \begin{pmatrix} a & c \\ b & 1 \end{pmatrix}$. It is given that

$$\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = x_1\begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} c \\ 1 \end{pmatrix} = x_2\begin{pmatrix} c \\ 1 \end{pmatrix} \text{ and } |P| = 1,$$

where *a*, *b* and *c* are constants.

(i) Find *P*.

(ii) Evaluate $P^{-1}\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}P$.

(iii) Using (b)(ii), evaluate $\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}^{12}$. ***(11 marks)***

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n^{th} Power of Square Matrix (by diagonization)

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HKAL Questions (Level 3)

1. ***[P.Math 2007 Paper I Q.5]***

Let P be a non-singular 2×2 real matrix and $Q=\begin{pmatrix}\alpha & 0\\ 0 & \beta\end{pmatrix}$, where α and β are two distinct real numbers. Define $M=P^{-1}QP$ and denote the 2×2 identity matrix by I.

(a) Find real numbers λ and μ, in terms of α and β, such that $M^2=\lambda M+\mu I$.

(b) Prove that $\det(M^2+\alpha\beta I)=\alpha\beta(\alpha+\beta)^2$. ***(6 marks)***

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HKAL Questions (Level 3)

2. ***[P.Math 1997 Paper I Q.7]***

(a) Let A be a 3×3 non-singular matrix. Show that

$$\det(A^{-1}-xI)=-\frac{x^3}{\det A}\det(A-x^{-1}I)$$

(b) Let $A=\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 4 & -17 & 8\end{pmatrix}$.

(i) Show that 4 is a root of $\det(A-xI)=0$ and hence find the other roots in surd form.

(ii) Solve $\det(A^{-1}-xI)=0$. ***(7 marks)***

HKDSE Questions (Level 3)

3. ***[M2 2014 Paper I Q.7]***

Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

(a) Prove that for all positive integers n, $A^{n+1} = 2^n A$.

(b) Using the result of (a), Willy proceeds in the following way:

$$A^2 = 2A$$
$$A^2 A^{-1} = 2AA^{-1}$$
$$A = 2I$$

Explain why Willy arrives at a wrong conclusion. ***(7 marks)***

HKDSE Questions (level 3)

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HKDSE Questions (Level 3)

4. ***[M2 2017 Q.12]***

Let $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$. Denote the 2×2 identity matrix by I.

(a) Using mathematical induction, prove that $A^n = 3^n I + 3^{n-1} n \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ for all positive integers n. ***(4 marks)***

(b) Let $B = \begin{pmatrix} 5 & 1 \\ -4 & 1 \end{pmatrix}$.

(i) Define $P = \begin{pmatrix} -1 & 0 \\ 2 & -1 \end{pmatrix}$. Evaluate $P^{-1}BP$.

(ii) Prove that $B^n = 3^n I + 3^{n-1} n \begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix}$ for any positive integer n.

(iii) Does there exist a positive integer m such that $\left|A^m - B^m\right| = 4m^2$? Explain your answer. ***(8 marks)***

HKDSE Questions (level 3)

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HKAL Questions (Level 4)

5. ***[P.Math 2011 Q.6]***

(a) Let $A = \begin{pmatrix} 4-b & a \\ b & 4-a \end{pmatrix}$ be a real matrix and $P = \begin{pmatrix} a & -1 \\ b & 1 \end{pmatrix}$, where $ab > 0$.

(i) Prove that P is a non-singular matrix.

(ii) Evaluate $P^{-1}AP$.

(iii) For any positive integer n, find d_1 and d_2 such that

$$A^n = P\begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix} P^{-1}.$$

(b) Let $B = \begin{pmatrix} 3 & 4 \\ 1 & 0 \end{pmatrix}$. For any positive integer n, find $B + B^3 + \ldots + B^{2n-1}$.

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HKAL Questions (Level 4)

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HKDSE Questions (Level 4)

6. ***[M2 2014 Q.12]***

Let $M = \begin{pmatrix} k-1 & k \\ 1 & 0 \end{pmatrix}$ and $A = \begin{pmatrix} 1 & p \\ -1 & 1 \end{pmatrix}$, where k and p are real numbers and $p \neq -1$.

(a) (i) Find A^{-1} in terms of p.

(ii) Show that $A^{-1}MA = \begin{pmatrix} -1 & k-p \\ 0 & k \end{pmatrix}$.

(iii) Suppose $p = k$. Using (ii), find M^n in terms of k and n, where n is a positive integer.

(8 marks)

(b) A sequence is defined by

$$x_1 = 0,\ x_2 = 1 \text{ and } x_n = x_{n-1} + 2x_{n-2} \text{ for } n = 3,4,5,\ldots .$$

It is known that this sequence can be expressed in the matrix form

$$\begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ x_{n-2} \end{pmatrix}.$$

Using the result of (a)(iii), express x_n in terms of n.

(3 marks)

HKDSE Questions (Level 4)

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HKAL Questions (Level 4)

7. ***[P.Math 1998 Paper I Q.9]***

Let $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \mathbf{R}$, $a \neq 0$ and $\det A = 0$.

(a) Show that $A=\begin{pmatrix} a & b \\ ka & kb \end{pmatrix}$ for some $k \in \mathbf{R}$. ***(3 marks)***

(b) Find P in the form of $\begin{pmatrix} 1 & 0 \\ r & 1 \end{pmatrix}$ such that $PA=\begin{pmatrix} \alpha & \beta \\ 0 & 0 \end{pmatrix}$ for some $\alpha, \beta \in \mathbf{R}$.

If $a+d \neq 0$, find Q in the form of $\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$ such that $PAP^{-1}Q=\begin{pmatrix} \gamma & 0 \\ 0 & 0 \end{pmatrix}$ for some $\gamma \in \mathbf{R}$.

(5 marks)

(c) Find S such that $S\begin{pmatrix} 3 & 7 \\ 6 & 14 \end{pmatrix}S^{-1}=\begin{pmatrix} \lambda & 0 \\ 0 & 0 \end{pmatrix}$ for some $\lambda \in \mathbf{R}$.

Hence, or otherwise, evaluate $\begin{pmatrix} 3 & 7 \\ 6 & 14 \end{pmatrix}^n$, where n is a positive integer.

(7 marks)

HKAL Questions (Level 4)

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HKAL Questions (Level 4)

8. ***[P.Math 2006 Paper I Q.8]***

Let $M = \begin{pmatrix} m & -m \\ m & m \end{pmatrix}$, where $m > 0$.

(a) Evaluate M^2. ***(1 mark)***

(b) Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a non-zero real matrix such that $MX = XM$.

(i) Prove that $c = -b$ and $d = a$.

(ii) Prove that X is a non-singular matrix.

(iii) Suppose that $X - 6X^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(1) Find X.

(2) If $a > 0$ and $(M - kX)^2 = -M^2$, express k in terms of m.

(10 marks)

(c) Using the result of (b)(iii)(2), find two real matrices P and Q, other than M and $-M$, such that $P^4 = Q^4 = M^4$. ***(4 marks)***

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HKAL Questions (Level 4)

Not for sale

HKAL Questions (Level 5)

9. ***[P.Math 2002 Paper I Q.12]***

(a) Let A be a 3×3 matrix such that $A^3+A^2+A+I=0$, where I is the 3×3 identity matrix.

(i) Prove that A has an inverse, and find A^{-1} in terms of A.

(ii) Prove that $A^4=I$.

(iii) Prove that $(A^{-1})^3+(A^{-1})^2+A^{-1}+I=0$

(iv) Find a 3×3 invertible matrix B such that $B^3+B^2+B+I\neq0$.

(6 marks)

(b) Let $X=\begin{pmatrix}1&1&1\\-1&-1&0\\-1&0&-1\end{pmatrix}$.

(i) Using (a)(*i*) or otherwise, find X^{-1}.

(ii) Let n be a positive integer. Find X^n.

(iii) Find two 3×3 matrices Y and Z, other than X, such that $Y^3+Y^2+Y+I=0$ and $Z^3+Z^2+Z+I=0$. ***(9 marks)***

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HKAL Questions (level 5)

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HKAL Questions (Level 5)

10. ***[P.Math 2005 Paper I Q.8]***

(a) Let $M = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, where $p, q, r, s \in \mathbf{R}$.

(i) Suppose det $M = 0$.
Prove that $M^{n+1} = (p+s)^n M$ for any positive integers n.

(ii) Suppose $qr > 0$. Let α and β be the roots of the quadratic equation $x^2 - (p+s)x + \det M = 0$
Denote the 2×2 identity matrix by I.

(1) Prove that α and β are two distinct real numbers.

(2) Prove that $M^2 - (\alpha+\beta)M + \alpha\beta I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

(3) Define $A = M - \alpha I$ and $B = M - \beta I$.
Prove that $AB = BA = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\det A = \det B = 0$.
Find real numbers λ and μ, in terms of α and β, such that $M = \lambda A + \mu B$.

(11 marks)

(b) Evaluate $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}^n$, where n is a positive integer.

(Hint: If X and Y satistfy $XY = YX = 0$, then $(X + Y)^n = X^n + Y^n$.)

(4 marks)

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HKAL Questions (level 5)

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HKDSE Questions (Level 5)

11. ***[M2 2013 Paper I Q.13]***

For any matrix $M=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, define $\text{tr}(M)=a+d$.

Let A and B be 2×2 matrices such that $BAB^{-1}=\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$.

(a) (i) For any matrix $N=\begin{pmatrix} e & f \\ g & h \end{pmatrix}$, prove that $\text{tr}(MN)=\text{tr}(NM)$.

(ii) Show that $\text{tr}(A)=4$.

(iii) Find the value of $|A|$.

(6 marks)

(b) Let $C=\begin{pmatrix} p & q \\ r & s \end{pmatrix}$. It is given that $C\begin{pmatrix} x \\ y \end{pmatrix}=\lambda_1\begin{pmatrix} x \\ y \end{pmatrix}$ and $C\begin{pmatrix} x \\ y \end{pmatrix}=\lambda_2\begin{pmatrix} x \\ y \end{pmatrix}$ for some non-zero matrices $\begin{pmatrix} x \\ y \end{pmatrix}$ and distinct scalars λ_1 and λ_2.

(i) Prove that $\begin{vmatrix} p-\lambda_1 & q \\ r & s-\lambda_1 \end{vmatrix}=0$ and $\begin{vmatrix} p-\lambda_2 & q \\ r & s-\lambda_2 \end{vmatrix}=0$.

(ii) Prove that λ_1 and λ_2 are the roots of the equation

$$\lambda^2-\text{tr}(C)\cdot\lambda+|C|=0 \ .$$

(5 marks)

(c) Find the two values of λ such that $A\begin{pmatrix} x \\ y \end{pmatrix}=\lambda\begin{pmatrix} x \\ y \end{pmatrix}$ for some non-zero matrices $\begin{pmatrix} x \\ y \end{pmatrix}$.

(2 marks)

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HKDSE Questions (Level 5)

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